Ans. (a) : The correct answer is (a) only 1 and 2.
Possible limitations of India in mitigating the global warming are lack of alternate technology to reduce emissions and being a developing country, India cannot invest huge amount of fund in research and development work. India itself contributes a lot in global warming due to its own industries, vehicles, less forest cover etc.

83. Consider the following which can be found in the ambient atmosphere:
1. Soot
2. Sulphur hexafluoride
3. Water vapour
Which of the above contribute to the warming up of the atmosphere?
(a) 1 and 2 only (b) 3 only
(c) 2 and 3 only (d) 1, 2 and 3

Ans. (d) : All above mentioned (soot, sulphur hexafluoride, water vapour) contribute to the warming up of the atmosphere.
Soot mostly from diesel engines is blocking snow and ice reflecting sunlight, which is contributing to near worldwide melting of ice and as much as a quarter of all observed global warming.
SF_6 (Sulphur hexafluoride) is the most potent as greenhouse gas that has been evaluated with a global warming potential of 23,900 times that is CO_2 when compared over a 100 years period.
Greenhouse effect is caused by atmospheric trace gases such as CO_2, O_3 (ozone) and water vapour (H_2O). Infrared radiation from the Earth is stored temporarily in the atmosphere. Of all these, trace gases, water vapour represents the most important constituent. It contributes to the natural greenhouse warming process by approximately 60%.

84. Consider the following statements:
1. The Taxus tree naturally found in the Himalayas.
2. The Taxus tree is listed in the Red Data Book.
3. A drug called "taxol" is obtained from Taxus tree is effective against Parkinson's disease.
Which of the statements given above is/are correct?
(a) 1 only (b) 1 and 2 only
(c) 2 and 3 only (d) 1, 2 and 3

Ans. (b) : Taxus Wallichiana Zucc (Taxus baccata L.) naturally occurs in the Himalayas. It has been included in the Red Data Book. The Taxol molecule, however, is very complex and therefore almost impossible to reproduce in the laboratory. One would have to rely on extractions of the Taxol compound the Taxus tree. Taxol is a cancer medication that interferes with the growth of cancer cells and slows their growth and spread in the body.
Taxol is used to treat breast cancer, lung cancer and ovarian cancer. It is also used to treat AIDS related Kaposi's sarcoma.

85. Due to their extensive rice cultivation, some regions may be contributing to global warming. To what possible reason/reasons is this attributable?
1. The anaerobic conditions associated with rice cultivation cause the emission of methane.
2. When nitrogen based fertilizers are used, nitrous oxide is emitted from the cultivated soil.
Which of the statements given above is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (c) : Methane and nitrous oxide are the two green house gases emitted from rice field. In rice field, the flooding a field cuts off the oxygen supply from atmosphere resulting anaerobic condition which emits methane and nitrogen fertilizer which generally used in rice field for high production, emits nitrous oxide by the reaction with atmospheric oxygen.

86. Sustainable development is described as the development that meets the needs of the present without compromising the ability of future generations to meet their own needs. In this perspective, inherently the concept of sustainable development is intertwined with which of the following concepts?
(a) Social justice and empowerment
(b) Inclusive Growth
(c) Globalization
(d) Carrying capacity

Ans. (d) : The correct answer is option (b).

87. Given below are the names of four energy crops. Which one of them can be cultivated for ethanol?
(a) Jatropha (b) Maize
(c) Pongamia (d) Sunflower

Ans. (b) : Maize can be cultivated for ethanol.

88. Some species of plants are insectivorous. Why?
(a) Their growth in shady and dark places does not allow them to undertake sufficient photosynthesis and thus they depend on insects for nutrition
(b) They are adopted to grow in nitrogen deficient soils and thus depend on insects for sufficient nitrogenous nutrition
(c) They cannot synthesize certain vitamins themselves and depend on the insects digested by them

(d) They have remained in that particular stage of evolution as living fossils, a link between autotrophs and heterotrophs

Ans. (b) : Insectivorous plants prey insects because most of the insectivorous plants grow in those area where there is deficit of nitrogen in nutrients so they catch insects to fulfill its need of nutrients.

89. Which one of the following reflects back more sunlight as compared to other three?
(a) Sand desert
(b) Paddy crop land
(c) Land covered with fresh snow
(d) Prairie land

Ans. (c) : Land covered with fresh snow reflects back more sunlight as compared to other three. Fresh snow and ice reflect 80 - 85% of the sunlight. Therefore, the skin of skiers is irradiated by sunlight from all sides, since the snow does not take up the light but reflects it and remains cold. Scientists give a name to this ability of surfaces to reflect light.

90. Which bacterial strain, developed from natural isolates by genetic manipulations, can be used for treating oil spills?
(a) Agrobacterium (b) Clostridium
(c) Nitrosomonas (d) Pseudomonas

Ans. (d) : Pseudomonas- bacterial strain, developed from natural isolates by genetic manipulations, can be used for treating oil spills.

91. Which feature of some species of blue green algae helps promote them as bio-fertilizers?
(a) They convert atmospheric methane into ammonia which the crop plants can absorb readily
(b) They induce the crop plants to produce the enzymes which help convert atmospheric nitrogen to nitrates
(c) They have the mechanism to convert atmospheric nitrogen into a form that the crop plants can absorb readily
(d) They induce the roots of the crop plants to absorb the soil nitrates in larger quantities

Ans. (c) : During the crop growth cycle, the blue-green algae grow, multiply, fix atmospheric nitrogen and make it available to the crop by way of excretion and autolysis.
Cyanobacteria, also known as blue-green algae (BGn). They have specialised cell i.e., heterocyst- It fixes nitrogen. BGn form a significant component of the marine nitrogen cycle and an important primary producer in many areas of the ocean, but are also found in habitats other than the marine environment.

92. India is a party to the Ramsar Convention and has declared many areas as Ramsar Sites. Which of the following statements best describes as to how we should maintain these sites in the context of this Convention?
(a) Keep all the sites completely inaccessible to man so that they will not be exploited
(b) Conserve all the sites through ecosystem approach and permit tourism and recreation only
(c) Conserve all the sites through ecosystem approach for a period without any exploitation, with specific period for each site, and then allow sustainable use of them by future generations.
(d) Conserve all the sites through ecosystem approach and allow their simultaneous sustainable use

Ans. (c) : Conserve all the sites through ecosystem approach for a period without any exploitation with specific period for each site, and then allow sustainable use of them by future generations.

93. Other than Jatropha curcas, why is Pongamia pinnata also considered a good option for the production of bio diesel in India?
1. Pongamia pinnata grows naturally in most of the arid regions of India
2. The sees of Pongamia pinnata are rich in lipid content of which nearly half is oleic acid.
Which of the statements given above is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (c)
⇒ Oil extracted from the Pongamia Pinnata seeds is the best substitute for kerosene for lighting lamps in rural areas.
⇒ The oil extracted has yet not used on a commercial level, but due to an increase in awareness and growth in research, Pongamia can be developed as an alternative source of fuel that can replace diesel from the market.
⇒ The oil extracted is commonly known as 'Karanja oil'.
⇒ Pongamia Pinnata is adaptable to arid zones, hence grows naturally in most of the arid regions of India. Hence statement 1 is correct.
⇒ Reasons for using Pongamia Pinnata as a source of biodiesel:
- It has a good nitrogen- fixing ability.
- The plant has the ability to grow in waterlogged, saline, and alkaline soil, wasteland and can withstand harsh agro climates.
- Pongamia Pinnata seeds contain 30% - 40% oil and little amount of free fatty acids. Monounsaturated and polyunsaturated fatty acids accounted for approx. 63% and 23%, respectively, of the seed oil and Oleic acid, is the major fatty acid found in the P. Pinnata seeds. Hence statement 2 is correct.

Hence option (c) is correct.